

Σίττυβος for Proofs in Elementary Linear Algebra

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1 Disclaimers

This is an experimental and independent course. Parts or all of the course may be terminated at any time and without notice.

Similarly, parts or all of the course materials, including the lecture recordings as well as written materials, may become unavailable at any time and without notice.

Engaging with the contents of this course is not a substitute for taking a comparable course that can be taken at an accredited university.

2 Staff and Meeting Hours

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Lectures: Asynchronous on Youtube at

[https://www.youtube.com/playlist?
list=PL40ydqvvyXfMwFmNvppWUuE3sgOP1RI5-](https://www.youtube.com/playlist?list=PL40ydqvvyXfMwFmNvppWUuE3sgOP1RI5-)

Course Webpage: As the course webpage we'll use the following Github repo:

[https://github.com/AlpUzman/
Proofs_in_Elementary_Linear_Algebra](https://github.com/AlpUzman/Proofs_in_Elementary_Linear_Algebra)

3 Course Description and Prerequisites

Proofs in Elementary Linear Algebra (PELA) is an independent, asynchronous and open access mini-course on proofs in elementary linear algebra. This mini-course is designed to address the transitory period from a typical introductory and computation-based linear algebra course to a proof-based linear algebra course. We'll focus on proof writing as a skill, both in terms of mathematics and in terms of presentation. Consequently we won't have time to cover mathematical topics that an ideal first proof-based mini-course in linear algebra ought to cover.

Throughout the lectures I will try to describe as clearly as possible my thought processes as I am writing proofs. I will follow as closely as possible the order in which topics are discussed in the main textbook (specified below in **Resources**), although I will not necessarily use the notation in the textbook. As a consequence of this, to better simulate how you might engage with the textbook the way it is presented, that is, in a linear fashion, we'll first cover all the definitions and theorems and then transition to solving the exercises. I will also separate finding the core idea of a proof from writing it as a rigorous piece of mathematical prose. Tentatively, I'll also separate writing a proof from polishing a written proof.

We'll first give a fast overview of common proof methods within the framework of the post-manifold hypothesis¹ and discuss the human and temporal nature of mathematical statements and proofs. This first part will be for general mathematical purposes; next we'll specialize to elementary linear algebra. While proof methods and many argumentation techniques are common to most mathematics, often the nature of a certain subfield of

¹The **post-manifold hypothesis** roughly can be described as follows: Mathematics is a dynamical thing in and of itself. It is structured constitutionally by mathematical truth. Mathematical truth is related to, but distinct from, many other truths like phenomenological truth, logical truth or physical truth. While mathematics as a post-manifold can be taken as a mathematical object, and hence is amenable to mathematical methods such as calculus (and downstream from it simplicial probing and coprobing) and dynamics (including shadowing), it can not be reduced to a purely mathematical object, nor can its constitution be separated from the consciousness of human mathematicians.

mathematics colors the actual proof writing regarding statements pertaining to that subfield. To keep the course self-contained and manageable, we'll focus only on some elementary results in linear algebra and their proofs, and we won't even mention eigenvalues for instance, presumably a topic that must be covered in any proof-based introduction to linear algebra. In the last part of the mini-course we'll transition to writing proofs as solutions to problems, as well as rewriting problems as solutions to problems.

Prerequisites There are no hard prerequisites for this mini-course, except some mathematical maturity. However familiarity with computation-based linear algebra ought to come in very handy. Ultimately this mini-course is mostly geared toward students who have taken an introductory linear algebra course and now starting a linear algebra course where they will be expected to prove statements.

Your Curator I obtained my PhD in mathematics from Penn State in 2023. From graduation until July 2026, I was a postdoc at the University of Utah. Since then I have been an independent educator and researcher (to use conventional words to approximately describe what I have been doing).

I was born and raised in Istanbul, Turkey, where I got my undergraduate education. There, I double majored in math and philosophy, and I learned how to communicate in English academically, how to write philosophy papers and how to write mathematical proofs. My perspective on proof writing has been shaped by this concurrent study, and broadly speaking, I think of mathematical writing to be in the same family as pretty much any other writing. This of course does not mean different kinds of writing don't have their own rules, including mathematics. Also, this does not mean in the world of mathematics the proofs-as-prose (or proofs-as-poetry, or proofs-as-performance) is the norm.

Bridging the Two Cultures For actually doing mathematics with other mathematicians, the culture so far has been mostly focused on efficiency and convenience, and in a more conventional math course I would de-emphasize many things we'll focus on in this mini-course (see e.g. my proof-based single-variable calculus courses: [2024 version](#) and [2026 version](#)). That said, with the latest developments downstream from A.I., it just might be the case that this culture will shift, bridging together the "[two cultures](#)" in a novel way. Perhaps due being an Istanbulite, I have a sensitivity toward perceiving potential bridges.

One of the consequences of the culture of efficiency and convenience that is not often talked about but one comes to learn from their peers and supervisors that a mathematical excerpt ought to be read only insofar as it helps with what the reader on their own private self are working on. This is reinforced with other politico-economical drivers like [publish-or-perish](#): only read what you think will be useful to you, because you do not (and ought not) have much time to do anything else. The communication skills of the mathematical writer who has become aware of the culture also degrades (or even never develops). There are exceptions as expected.

Based on my own experience as an educator of proof-based classes, I also have the sense that the culture of efficiency and convenience when it comes to communication alienates many students from mathematics, who would have otherwise been very interested. This consequently results in us mathematicians losing potential unique perspectives.

Experimentality I hope the fundamentally experimental nature of this mini-course is fairly apparent. Earlier this year, I was planning to give an independent, year-long, live-streamed, open-access course on some math+topic I ultimately did not announce, but upon further consideration [I decided to postpone](#). One of the reasons was the sheer amount of moving parts I am not familiar with. So I decided to factorize this larger project, and the design of this mini-course is built around some of the factors. It's much

shorter in length, on a topic that is more conventional, and asynchronous. This will help me find my way through course and set design, lighting, camera work, post-production as well as cybernetic dissemination and the associated coaching aspect (more on this below in [Ways to Support](#)).

This mini-course is also cognate with my course-fiction, "[Future-Proving for Math Majors](#)", and while many ideas between the two resonate, they each can be engaged with as standalone complexes.

4 Resources

For this mini-course we'll follow the 17th chapter ("Proofs in Linear Algebra") of the following textbook:

Chartrand, G.; Polimeni, Albert D.; Zhang, P. *Mathematical Proofs: A Transition to Advanced Mathematics*, 4e. Pearson.
2018. ISBN: 9780134746753

This chapter specifically is freely available through the publisher's webpage at

https://media.pearsoncmg.com/cm/pmmg_mml_shared/Chartrand_4e/index.html

5 Tentative Schedule

The course's subject matter is divided into episodes as follows. For your convenience the approximate corresponding sections in the main textbook as well as supplementary materials are also listed.

Part I: Mathematical Statements and Their Proofs

Episode 1: Σίττυβος Day

Episode 2: Proof Methods Post-Manifold Hypothesis. Naive categorical logic. Logic: Statements. Universally True Statement. Universally False Statement. Logical Operators: and, or, xor, not, implies. Logical Quantifiers: for all, there exists, there uniquely exists. Proof Methods: direct proof, proof by contrapositive, proof by contradiction, proof by induction, case analysis, wlog, A.I..

Episode 3: Phenomenology Definition of phenomenology. Famous phenomenologists. Phenomenological Triangle. Indexing (aka ἐποχή ("epoche") aka bracketing aka transcendental reduction). Perturbation (aka eidetic variation). Statical versus dynamical phenomenology (aka descriptive versus genetic phenomenology). Phenomenologically informed mathematics.

Part II: Theorems in Elementary Linear Algebra and Their Proofs

Episode 4: Vector Space Axiomatics (§17.1-4) Vector space. Vector. Scalar. Linear combination. Zero vector. Negative of a vector. Function spaces. Polynomial spaces. Matrix spaces. Matrix-Matrix product. Inner product.

Episode 5: Subspaces, Spans, Linear Independence (§17.5-7) Closure under addition. Closure under scalar multiplication. Vector subspace. Span. Linear independence. Linear dependence.

Episode 6: Linear Transformations (§17.8-9) Vector addition preservation. Scalar multiplication preservation. Linear transformation. Domain. Codomain. Image. Preimage. Kernel.

Part III: Proofs and Problems as Solutions to Problems

Episode 7: Problem Solving

Episode 8: Problem Writing

6 Lectures

Lectures will be asynchronous. Edited and annotated lecture recordings will be uploaded regularly (approximately one episode per two weeks) to the following YouTube playlist:

[https://www.youtube.com/playlist?
list=PL40ydqvvyXfMwFmNvppWUuE3sgOP1RI5-](https://www.youtube.com/playlist?list=PL40ydqvvyXfMwFmNvppWUuE3sgOP1RI5-)

7 Learning Objectives

Among the objectives in this mini-course are that you learn how to:

- use logical quantifiers,
- parse mathematical statements,
- distinguish levels of rigor in mathematical prose,
- construct and write proofs in accordance with modern standards of mathematical rigor,
- construct and write mathematical problems that are consistent and solvable relative to the allowed resources,

(all within the context of elementary linear algebra).

8 Office Hours

There will be no office hours. However you may seek help on public platforms; I will pay extra attention to comments left on Youtube under the videos.

Another, more private, option is to reach out via an email to

uzmancoaching@gmail.com.

Make sure to include "PELA " in the subject line, and be as clear and detailed as possible with your query. Remember: you would be contacting me this way because you want my own personal opinion or feedback, as opposed to, say, asking your question to A.I.. Unless you have hired me as a coach, I may or may not answer your first private query.

9 Problem Sets

There will be no regularly released problem sets for this mini-course. However in the last part of the mini-course we'll focus specifically on problem writing, and we'll create one problem set, which you may then use as practice, with or without referring to the rest of the mini-course.

10 Exams and Grades

There will be no examinations, and there will be no grading.

11 Conduct

You are expected to act in accordance with the terms of service of platforms this mini-course will be using. I also expect you to interact with the course content with the awareness and understanding that public platforms are public spaces, despite the fact that you might be, or think you are, anonymous.

12 Ways to Support

This online mini-course is asynchronous and independent. You may support me through the following means:

Eyes: I need more eyes on this project as well as the previous work I have done. To help out with this, you can act in ways that would generate an appropriate feedback loop in the **cybernetic** system made up of the platforms used to deliver the content. More succinctly, this means commenting on and liking the videos, subscribing to the channel, and sharing the videos and other supplementary material.

While I currently have no institution affiliation, I am not opposed to an affiliation and I will likely be applying to jobs in academia (as of late September 2026). In particular, I would very much appreciate the opportunity to be able to teach a course like this at a university, and if you are a student at a university, mentioning this mini-course to your professors might help me in this regard.

Time: I need surplus time to spend on producing content like this. As time itself is not really transferable from person to person, the way to help out in this regard is through providing financial support. Currently the only option available is subscribing for a Youtube membership to my channel, through the following link:

<https://www.youtube.com/channel/UCcod9rszdguv5QYIDpJfo3A/join>.

Private Coaching Send an email to

uzmancoaching@gmail.com

if you are interested in private coaching, including personalized feedback on your work on course materials (exercises given during lectures, problem sets and exams). Make sure to include your mathematical level and why you are following this course in the body of the email. Make sure also to include "PELA " in the subject line.

13 License

All contents of this mini-course (except the chapter we'll be following, whose copyright belongs to Pearson Education, Inc.) are licensed under CC BY-NC-SA 4.0 (Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International Public License); for further details see:

<https://creativecommons.org/licenses/by-nc-sa/4.0/>

14 Acknowledgements

This mini-course can be considered as pedagogical action; big thanks to P.F.. While I am not against the use of A.I. in general, I did not explicitly use A.I. in retouching this σίττυβος .